

**ShanghaiTech University**  
**School of Information Science and Technology**

**Lab Experiments**

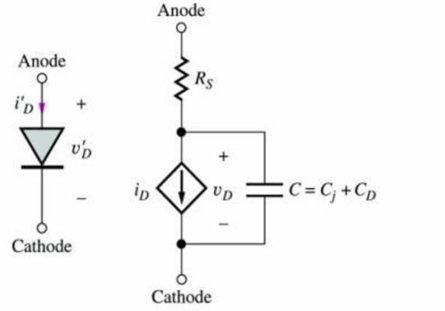
**Diode**

**PN Junction:**

In class, we described the current-voltage relationship as

$$i_D = I_s \left[ \exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad (1)$$

The parameter  $n$  is called “ideality factor”. For ideal diode at normal current,  $n$  is equal to 1. In real devices, it could deviate from 1. The **ideality factor** can be easily extracted from the semi-log plot of  $i_D$  vs.  $v_D$ . The SPICE model of a pn junction is shown below:



The SPICE model also includes a series resistance,  $R_S$ . Its value can be extracted from the slope of the I-V curve at high current: the total resistance at high current is dominated by  $R_S$ .

The abrupt PN junction (This corresponds to pn junctions with constant doping concentrations, i.e.,  $N_D$  and  $N_A$  are constant.) looks like a pair of parallel plates of area  $A$ , separation  $W$ , and permittivity  $\epsilon$ . This capacitance is given by,

$$C = \frac{A\epsilon}{W} \quad (2)$$

However, the width of the depletion (junction) region of a PN junction is not constant.  $W$  varies with the dc voltage across the junction as

$$W(V) = (V_0 - V)^{1/2} \left( \frac{2\epsilon_r\epsilon_0}{q} \right)^{1/2} \left( \frac{N_a + N_d}{N_a N_d} \right)^{1/2} \quad (3)$$

where  $V_0$  is the built-in potential of the device (a constant). The junction capacitance then becomes:

$$C_j(V) = A \left( \frac{q\epsilon_r\epsilon_0}{2} \frac{N_a N_d}{N_a + N_d} \right)^{1/2} \frac{1}{|V_0 - V|^{1/2}} \quad (4)$$

where  $V$  is the dc applied voltage. The junction may be forward biased ( $V > 0$ ) or reverse biased ( $V < 0$ ) but Eq. (4) is the total capacitance only for reverse bias. If we square equation Eq. (4) and rewrite it, we have,

$$\frac{1}{C_j^2} = K(V_0 - V) \quad (5)$$

where  $K = [(N_a + N_d)/(N_a * N_d)] * 2 / (q * \epsilon_r * \epsilon_0 * A^2)$ . A plot of  $1/C_j^2$  vs.  $V$  as measured should be a straight line for the ideal abrupt PN junction. The **built-in voltage  $V_0$**  can be extracted from experimental data by curve fitting.